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1 Compactness

We will now discuss the equivalence of topological and sequential compactness in the reals and in general metric spaces. Let us view the following theorem:

1.1 Sequential Compactness

Theorem 1.1 (Sequential Bolzano Weierstrass). *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

Proof. Let us define the following set, we call "peaks" of a sequence:

$$P := \{n \in \mathbb{N} : (\forall j > n : x_j < x_n)\}$$

In the case that P is countably infinite, then we set:

$$n_0 := \inf P \wedge \forall k > 0 : n_k := \inf (P \setminus \{n_0, \dots, n_{k-1}\})$$

In which case we are guaranteed that $\forall k \in \mathbb{N} : n_{k+1} > n_k$ which tells us that $x_{n_{k+1}} < x_{n_k}$, showing that there exists a strictly decreasing subsequence. In the case that P is finite, we choose the following:

$$n_0 := \sup(P) + 1$$

that way the peaks are fully omitted from the subsequence, as they will prove to be problematic in the construction. $\square$